2) a)

$\frac{1}{s} \to \boxed{\frac{5}{s+5}} \to C(s)$

$C(s) = \frac{1}{s} \cdot \frac{5}{s+5} = \frac{5}{s(s+5)}$

$\frac{5}{s(s+5)} = \frac{A}{s} + \frac{B}{s+5}$ $\quad 5 = A(s+5) + Bs$ $\quad s = 0;\ A = 1$; $s = -5;\ B = -1$

$C(s) = \frac{1}{s} + \frac{-1}{s+5} \Rightarrow C(t) = 1 + (-e^{-5t}) = 1 - e^{-5t}$

$\tau = 1/5\ s$ $\quad T_r = 2.2/5\ s$ $\quad T_s = 4/5\ s$

b)

$\frac{1}{s} \to \boxed{\frac{20}{s+20}} \to C(s)$

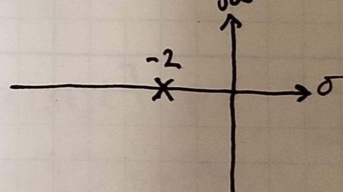
$C(s) = \frac{20}{s(s+20)}$

$\frac{20}{s(s+20)} = \frac{A}{s} + \frac{B}{s+20}$ $\quad 20 = A(s+20) + Bs$ $\quad s = 0;\ A = 1$; $s = -20;\ B = -1$

$C(s) = \frac{1}{s} + \frac{-1}{s+20}$ $\quad C(t) = 1 -$

$\tau = 1/20\ s$ $\quad T_r = 2.2/20\ s$ $\quad T_s = 4/20\ s$

8) a) $T(s) = \frac{2}{s+2}$ $\quad$ root $= -2$



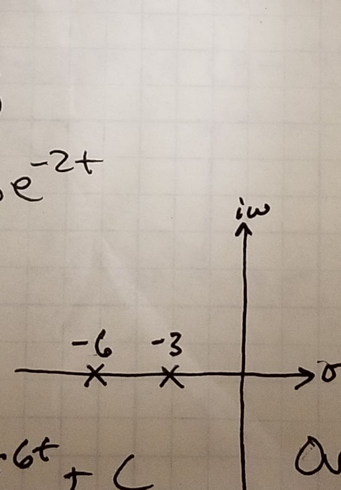
step-response

$T(s) = \frac{2}{s(s+2)}$

Overdamped

$C(t) = A + Be^{-2t}$

b) $T(s) = \frac{5}{(s+3)(s+6)}$



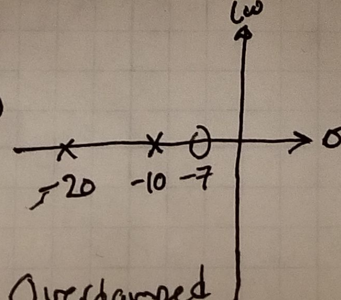
Step-response

$T(s) = \frac{5}{s(s+3)(s+6)}$

$C(t) = Ae^{-3t} + Be^{-6t} + C$

Overdamped

c) $T(s) = \frac{10(s+7)}{(s+10)(s+20)}$



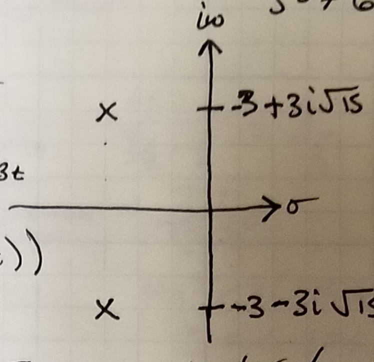
$C(s) = \frac{10(s+7)}{s(s+10)(s+20)}$

$C(t) = A + Be^{-10t} + Ce^{-20t}$

Overdamped

d) $T(s) = \dfrac{20}{s^2+6s+144} \rightarrow$ step response $C(s) = \dfrac{20}{s^3+6s^2+144s}$

$= \dfrac{20}{s(s+3-3i\sqrt{15})(s+3+3i\sqrt{15})}$

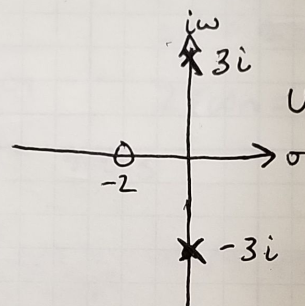


under damped

$C(t) = A + (B\cos(3\sqrt{15}t) + Bi\sin(3\sqrt{15}t)e^{-3t}$
$+ e^{-3t}(C\cos(3\sqrt{15}t) - Ci\sin(3\sqrt{15}t))$

$C(t) = A + e^{-3t}[\overset{D}{(B+C)}\cos(3\sqrt{15}t)$
$+ \underset{E}{(B-C)}i\sin(3\sqrt{15}t)]$

e) $T(s) = \dfrac{s+2}{s^2+9}$



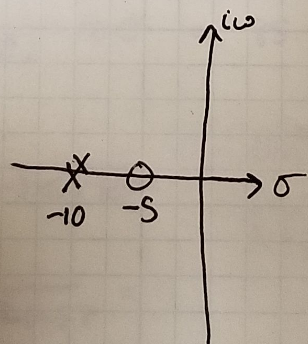
Undamped

$C(s) = \dfrac{s+2}{s(s^2+9)}$

$C(t) = A + e^{-t\cdot 0}[B\cos(3t) + B\sin(3t)i] + e^{-t\cdot 0}[C\cos(3t) - Ci\sin(3t)]$

$C(t) = A + (B+C)\cos(3t) + (B-C)i\sin(3t)$

f) $T(s) = \dfrac{s+5}{(s+10)^2}$



Step response

$C(s) = \dfrac{s+5}{s(s+10)^2}$

$C(t) = A + Be^{-10t} + Ce^{-10t}t$

20) a)

$$T(s) = \frac{16}{s^2+3s+16} \qquad \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2} = T$$

$$\omega_n^2 = 16 \rightarrow \omega_n = 4$$

$$2\omega_n\zeta = 3 \rightarrow \zeta = 3/8 \qquad T_s = \frac{4}{\zeta\omega_n} = \frac{4}{3/8 \cdot 4} = \frac{8}{3}$$

$$T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{4\sqrt{1-9/64}} = \frac{\pi}{4\sqrt{55/64}} = \frac{2\pi}{\sqrt{55}}$$

$$\%OS = \exp\left(\frac{\zeta\pi}{\sqrt{1-\zeta^2}}\right)\cdot 100 \qquad \omega_n T_r = (1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1)$$

$$\%OS = \exp\left(\frac{3/8\,\pi}{\sqrt{1-3/8^2}}\right)\cdot 100 = 28.06\% \qquad T_r = 0.356$$

b) $$T(s) = \frac{0.04}{s^2+0.02s+0.04} \qquad \omega_n^2 = 0.04 \qquad \omega_n = 0.2$$

$$2\zeta\omega_n = 0.02 \rightarrow \zeta = 0.05$$

$$T_s = \frac{4}{\zeta\omega_n} = \frac{4}{0.2\cdot 0.05} = 400\text{ s}$$

$$T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{0.2\sqrt{1-0.05^2}} = 15.728$$

$$T_r = \frac{(1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1)}{\omega_n} = 5.26\text{ s}$$

$$\%OS = \exp\left(-\frac{0.05\pi}{\sqrt{1-0.05^2}}\right)\cdot 100 = 85.45\%$$

c) $$T(s) = \frac{1.05\times10^7}{s^2+1.6\times10^3 s+1.05\times10^7} \qquad \omega_n^2 = 1.05\times10^7 \qquad \omega_n = 3240.37$$

$$2\zeta\omega_n = 1.6\times10^3 \rightarrow \zeta = 0.247$$

$$T_s = \frac{4}{\zeta\omega_n} = \frac{4}{0.247\cdot 3240.37} =$$

$$T_p = \frac{\pi}{3240.37\sqrt{1-0.247^2}} = 0.001$$

$$T_r = \frac{(1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1)}{\omega_n} = 0.000388$$

$$\%OS = \exp\left(-\zeta\pi/\sqrt{1-\zeta^2}\right)\cdot 100 = 44.92\%$$

23) a) $T_s = \frac{4}{\zeta \omega_n}$

$$\%OS = \exp\left(-\frac{\zeta \pi}{\sqrt{1-\zeta^2}}\right) \cdot 100$$

$$\ln\left[\frac{\%OS}{100}\right] = -\frac{\zeta \pi}{\sqrt{1-\zeta^2}}$$

$$\sqrt{1-\zeta^2}\,\ln\left[\frac{\%OS}{100}\right] = -\zeta\pi$$

$$(1-\zeta^2)\ln\left[\frac{\%OS}{100}\right]^2 = \zeta^2\pi^2$$

$$\cancel{\ln\left[\frac{\%OS}{100}\right]^2 - \zeta^2 \ln\left[\frac{\%OS}{100}\right]^2 = \zeta^2\pi^2}$$

$$\cancel{\ln\left[\frac{\%OS}{100}\right] = \zeta^2\left(\pi^2 + \ln\left[\frac{\%OS}{100}\right]\right)}$$

$$\cancel{\zeta = \frac{\ln\left[\frac{\%OS}{100}\right]^2}{\pi^2 + \ln\left[\frac{\%OS}{100}\right]^2}}$$

$$\zeta = \frac{2\ln\left[\frac{\%OS}{100}\right]}{\pi^2 + 2\ln\left[\frac{\%OS}{100}\right]}$$

$$\zeta =$$

$$OS = e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}}$$

$$\ln[OS] = -\frac{\zeta\pi}{\sqrt{1-\zeta^2}}$$

$$\sqrt{1-\zeta^2}\,\ln[OS] = -\zeta\pi$$

$$\ln[OS]^2 - \zeta^2\ln[OS]^2 = (-\zeta\pi)^2$$

$$\ln[OS] = \zeta^2\pi^2 + \zeta^2\ln[OS]$$

$$\ln[OS] = \zeta^2[\pi^2 + \ln[OS]^2]$$

$$\frac{\ln[OS]^2}{\pi^2 + \ln[OS]^2} = \zeta^2$$

$\zeta = 0.559$

$$T_s = \frac{4}{0.559\,\omega_n} \rightarrow \omega_n = \frac{4}{0.559 \cdot 0.6}$$

$$\omega_n = 11.926$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$

poles $= -6.667 \pm 9.878i$

23) b) $\%OS = 10\%$ $\quad T_p = 5$ sec

$$\zeta = \left[\frac{\ln[OS/100]^2}{\pi^2 + \ln[OS/100]^2}\right]^{1/2} = 0.5912$$

$$\omega_n = \frac{\pi}{T_p\sqrt{1-\zeta^2}} = 0.779$$

$$s^2 + 2\zeta\omega_n + \omega_n^2 = 0 \qquad \text{roots} = -0.4605 \pm 0.6283i$$

c) $T_s = 7 \quad T_p = 3$

$$\zeta\omega_n = 4/T_s = 4/7$$

$$\omega_n\sqrt{1-\zeta^2} = \pi/T_p = \pi/3 \qquad \text{roots} = -4/7 \pm \pi/3\, i$$

24) $\%OS = 15\%$ $\quad T_s = 0.7$ sec

$$\zeta = \left[\frac{\ln[OS/100]^2}{\pi^2 + \ln[OS/100]^2}\right]^{1/2} = 0.517$$

$$\omega_n = \frac{4}{\zeta \cdot T_s} = 11.054$$

$$\text{Transfer Function} = \frac{\omega_n^2}{s^2 + 2\omega_n\zeta + \omega_n^2} = \frac{0.267}{s^2 + 11.429s + 122.197}$$

25) $5\ddot{x} + 2\dot{x} + 20x = f(t)$

$X(s)(5s^2 + 2s + 20) = F(s)$

$X(s) = \frac{F(s)}{5s^2 + 2s + 20}$ $\qquad \frac{X(s)}{F(s)} = \frac{1}{5s^2+2s+20} = \frac{1/5}{s^2 + 2/5 s + 4}$

$\omega_n^2 = 4 \rightarrow \omega_n = 2$ $\qquad \%OS = \exp\left(-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}\right) \cdot 100 = 72.925\%$

$2\zeta\omega_n = 2/5 \rightarrow \zeta = 1/10$

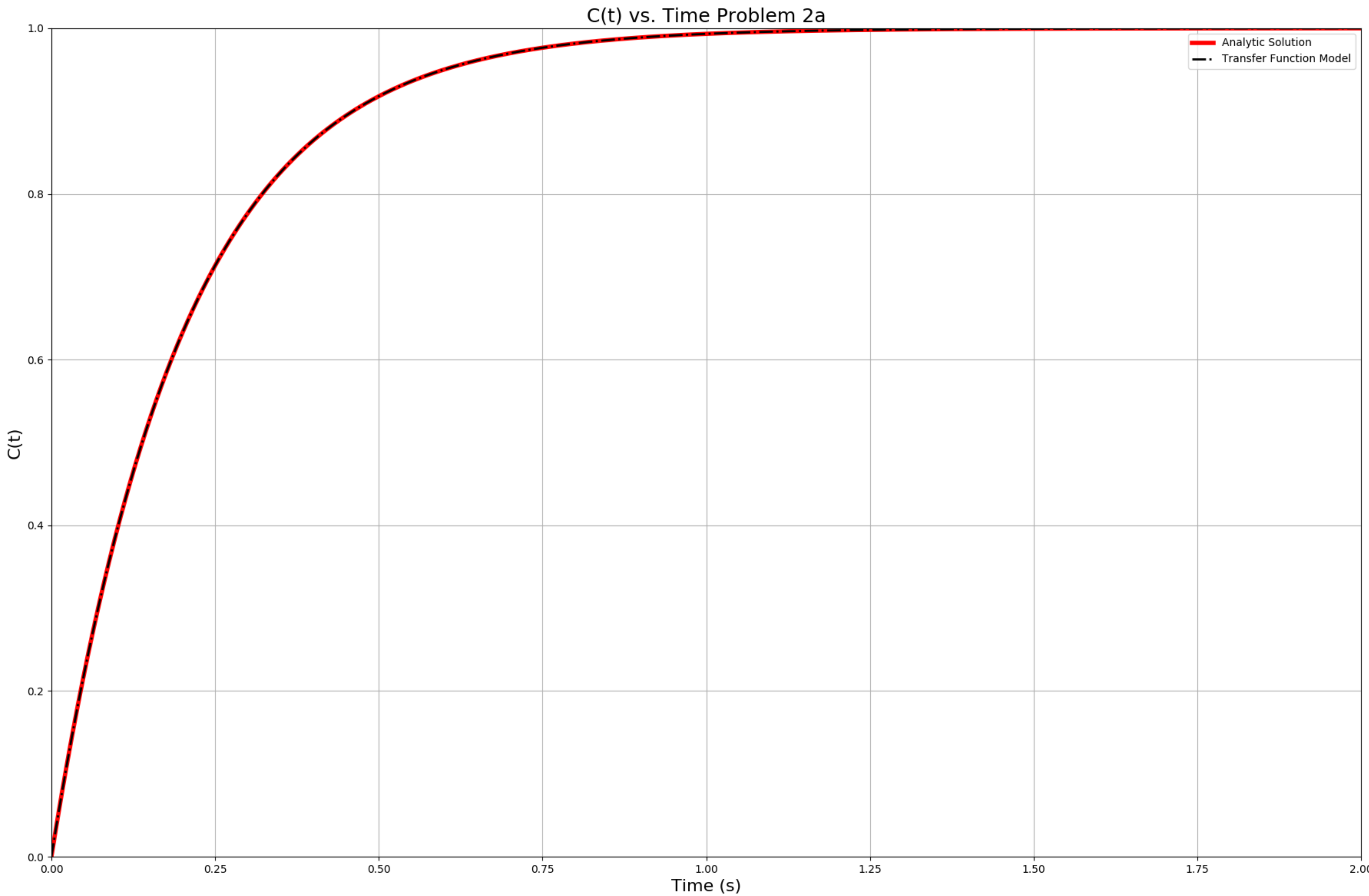
$T_s = \frac{4}{\frac{1}{10} \cdot 2} = 20 \text{ sec}$ $\qquad T_p = \frac{\pi}{2 \cdot \sqrt{1 - 2/5^2}} = 1.579 \text{ sec}$

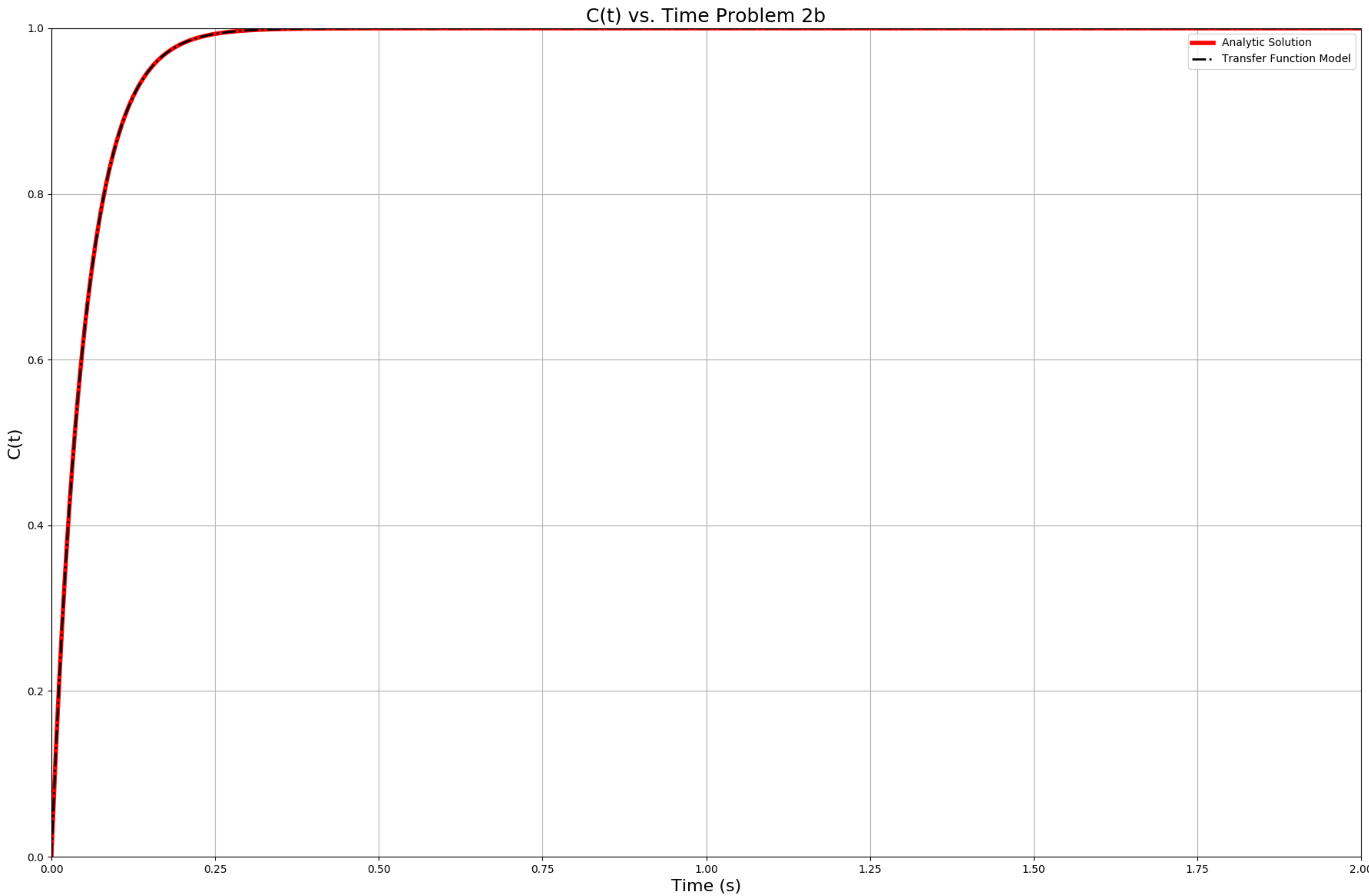
$T_r = \frac{1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1}{\omega_n} = 0.551 s$

$C_{final} = \frac{0.2}{s^2 + 0.4s + 4} = \frac{0.2}{4} = 0.05$

```python
# -*- coding: utf-8 -*-
"""
Created on Wed Sep 26 21:34:35 2018

@author: jfowl
"""
import numpy as np
import control as ctl
import matplotlib.pyplot as plt

this_clss = 'ME_426'
homework = 'HW_6'



# %% PROBLEM 2a FOR PROBLEM 3


def sys_model2a(tv):
    """Problem 2a"""
    x = []
    for t in tv:
        x.append(1 - np.exp(-5*t))
    return x


tvec2a = np.linspace(0, 2, 1000)
tf2a = ctl.tf([5], [1, 5, 0])
tout2a, cout2a = ctl.impulse_response(tf2a, tvec2a)

plt.figure(1, figsize=(22, 14))
plt.plot(tvec2a, sys_model2a(tvec2a), 'r', linewidth=4,
         label='Analytic Solution')
plt.plot(tout2a, cout2a, '-.k', linewidth=2, label='Transfer Function Model')
plt.xlim([min(tvec2a), max(tvec2a)])
plt.ylim([min(cout2a), max(cout2a)])
plt.xlabel('Time (s)', fontsize=16)
plt.ylabel('C(t)', fontsize=16)
plt.title('C(t) vs. Time {}'.format(sys_model2a.__doc__), fontsize=18)
plt.legend()
plt.grid()
plt.savefig('{}_{}_{}_Plot.png'.format(this_clss, homework,
            sys_model2a.__doc__))
plt.show()
# %% PROBLEM 2b FOR PROBLEM 3


def sys_model2b(tv):
    """Problem 2b"""
    x = []
    for t in tv:
        x.append(1 - np.exp(-20*t))
    return x


tvec2b = np.linspace(0, 2, 1000)
tf2b = ctl.tf([20], [1, 20, 0])
```

```
tout2b, cout2b = ctl.impulse_response(tf2b, tvec2b)

plt.figure(2, figsize=(22, 14))
plt.plot(tvec2b, sys_model2b(tvec2b), 'r', linewidth=4,
         label='Analytic Solution')
plt.plot(tout2b, cout2b, '-.k', linewidth=2, label='Transfer Function Model')
plt.xlim([min(tvec2b), max(tvec2b)])
plt.ylim([min(cout2b), max(cout2b)])
plt.xlabel('Time (s)', fontsize=16)
plt.ylabel('C(t)', fontsize=16)
plt.title('C(t) vs. Time {}'.format(sys_model2b.__doc__), fontsize=18)
plt.legend()
plt.grid()
plt.savefig('{}_{}_{}_Plot.png'.format(this_clss, homework,
            sys_model2b.__doc__))
plt.show()
```


C(t) vs. Time Problem 2a
Analytic Solution
Transfer Function Model
C(t)
Time (s)
0.0
0.2
0.4
0.6
0.8
1.0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00


C(t) vs. Time Problem 2b
Analytic Solution
Transfer Function Model
C(t)
Time (s)
0.0
0.2
0.4
0.6
0.8
1.0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00